

← ↻ localhost:8501

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Deploy ⋮

How detection works

Stage 1 – XGBoost Supervised model trained on labeled web attacks. High precision on known patterns.

Stage 2 – Isolation Forest Unsupervised anomaly detector. Flags flows that don't look like normal traffic – a safety net for novel attacks.

Combined = semi-supervised. A flow is flagged if *either* stage flags it (maximizes recall).

Detection threshold

XGBoost attack probability

0.50

0.001.00

CIC-IDS-2017 · SEMI-SUPERVISED IDS

◆ SENTINEL — Web Attack Detection Console

XGBoost (supervised) + Isolation Forest (anomaly) - explained with SHAP

◆ Analyze a flow ◆ Batch scan (CSV) ◆ Sequence scan (CRP) ◆ About

Upload a single flow to analyze

Upload a one-row CSV (a single network flow with the CICFlowMeter feature columns). The console runs both detectors and shows a verdict.

Single-flow CSV

Drag and drop file here

Limit 200MB per file • CSV

Browse files

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0.50

0.001.00

✓ BENIGN — no threat

Both detectors consider this flow normal.

0.0%
XGBOOST ATTACK PROB.

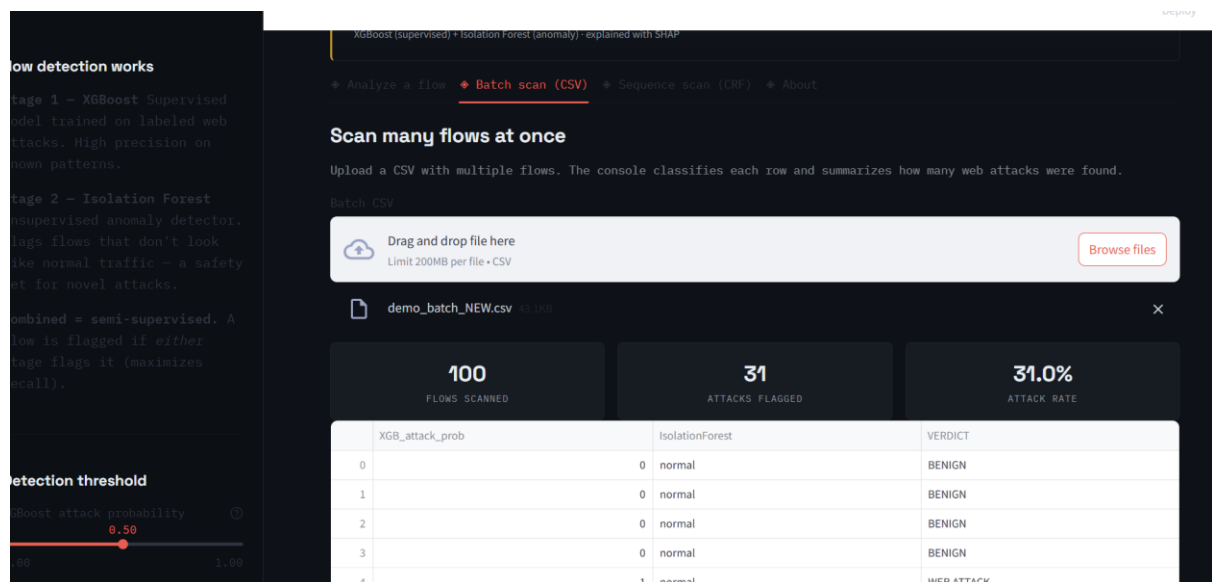
NORMAL
ISOLATION FOREST

BENIGN
COMBINED VERDICT

Why this verdict? (SHAP)

Top features driving this decision

Init_Win_bytes_backward	
Init_Win_bytes_forward	
Flow IAT Min	
URG Flag Count	
Flow IAT Mean	
Max Packet Length	
Fwd IAT Min	
Flow IAT Std	
Flow Duration	



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Detection threshold

XGBoost attack probability

0.50 0.50 1.00

The CRF is a sequence model: it needs MANY flows in time order and labels each flow using its neighbours. Upload a batch CSV (e.g. demo_batch_NEW.csv) with the feature columns, in capture order.

Sequence CSV (many flows, in order)



Drag and drop file here

Limit 200MB per file • CSV

Browse files



demo_crf_sequence.csv 0.6MB



1500

Flows in sequence

46

Flagged by CRF

